
Toward Mechanical Scientific Models: Fractal Custody and Governed Agent Experiments

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Abstract

1 The *Mechanical Scientific Method* treats scientific computation as a sequence of
2 governed state transitions in which evidence, transforms, and bounded claims re-
3 main inspectable rather than collapsed into model prose. We propose the term
4 *Mechanical Scientific Model* for a computational model class that satisfies explicit
5 requirements: explicit evidence state; governed and versioned state transitions;
6 separation of deterministic transforms from probabilistic model outputs; preserva-
7 tion of positive, null, negative, failed, blocked, timeout, abstention, and contradic-
8 tory outcomes; explicit claim ceilings; forward and reverse evidence traceability;
9 and successor-state recording rather than silent history rewriting. *Fractal Custody*
10 *Objects* (FCOs) and a *Fractal Custody Graph* (FCG) provide the custody
11 substrate for binding those transitions. HydraDG is the primary experimentally
12 evaluated implementation examined here: it preregisters hypotheses, freezes case
13 manifests and scorers, records raw prompt and response hashes, and preserves
14 terminal states without silent substitution. We report two preregistered HydraDG
15 studies (EXP-008, EXP-009) comparing flat prose versus structured FCG retrieval
16 for failure-prevention endpoints; both terminate as *underpowered*, so confirmatory
17 treatment-effect claims are not supported while null, negative, and failure evidence
18 remains in custody. We do not claim that Mechanical Scientific Models are an es-
19 tablished external field term; we offer them as a proposed model class and report
20 bounded custody behavior from one implementation.

21 1 Introduction

22 Large language model agents are evaluated on task success, benchmark scores, and human prefer-
23 ence [2, 3]. Less often preserved are the conditions under which experiments fail: swapped models,
24 stale runtime selection, partial matrix execution, or JSON outputs that never parse. When negative
25 and null cells disappear from the public record, later work cannot distinguish *no effect* from *no*
26 *measurement*.

27 This paper separates five layers that are often conflated in agent-evaluation reporting:

- 28 • **Method** — the *Mechanical Scientific Method*: governed transitions among explicit evi-
29 dence states rather than narrative collapse of runs into conclusions.
- 30 • **Model class** — *Mechanical Scientific Models* (proposed here; not an established external
31 terminology): computational systems whose state transitions are mechanically exposable
32 and failure-complete.
- 33 • **Custody substrate** — FCO/FCG receipts that bind source bytes, transforms, derived evi-
34 dence, bounded claims, and successor state.

- **Primary implementation** — HydraDG, the experimentally evaluated implementation whose preregistered EXP-008 and EXP-009 results are reported here.
- **Systems validation and companion exemplars** — HydraLamp perturbation receipts and HydraDG scoring on local Apple Silicon (Mac Studio class, 32 GB RAM); Antigence anticube lanes (supplement F11–F12); Vithia companion lineage in repositories vitaology and fractal-custody-objects (ZERO_PRIMARY_WEIGHT per cross-repo census; companion runs not vendored into this submission checkout).

HydraDG instantiates the proposed model class through a custody-first experimental contract. Every actor—human operator, frontier agent, local model runtime, or deterministic script—must emit or be represented by a handoff receipt before its output is promoted. Receipts bind actor identity, host, repository state, input hashes, raw output hashes, evidence class, claim ceiling, and signature state (hashes are identity commitments, not digital signatures unless an authorized signing operation occurred). Probabilistic model outputs are never treated as authority; they are evidence objects subject to deterministic scoring and graph append rules.

This paper reports terminal results from the IC Failure Learning structured-retrieval falsification lane executed in HydraDG. Primary numbers are reverse-traced to frozen preregistered verdict JSON receipts; companion tables and figures cite SHA-256-bound supplement artifacts bundled with this submission. We do *not* claim that structured FCG context improves failure prevention: EXP-008 and EXP-009 closed as *underpowered* (`n_paired= 2`, `discordant= 0`) with `conclusion_bounded=effect not established`. Exploratory secondary patterns and vitaology runs are not promoted to primary findings for those experiments.

2 Related Work

Agent evaluation. Benchmark suites measure capability but rarely require publication of failed runs, malformed generations, or infrastructure blocks [2, 3]. HydraDG complements benchmarks with mandatory negative capture at the custody layer.

Retrieval and structured context. Retrieval-augmented generation and graph-augmented memory systems assume structured context helps [1, 4]. Falsification requires frozen contracts, explicit null retention, and power-aware verdicts; HydraDG enforces both.

Local open-weights evaluation. Frozen EXP-008/009 cells use qwen3:1.7b and qwen2.5-coder:7b only; LongMemEval track-03 rows in supplement table A4 use qwen2.5-7b and deepseek-r1-14b from eval/track_model_k_20260820/ with claim ceiling HISTORICAL_SUPPORTING_ZERO_PRIMARY_WEIGHT. No frontier API model executed a preregistered primary scorer in this submission.

Provenance, integrity, and reproducibility. W3C PROV-O and workflow provenance systems represent process and research-object lineage [12, 13], while append-only Merkle audit logs provide established tamper-evident integrity mechanisms [11]; HydraDG does not claim to invent these primitives. Recent work surveys evidence tracing and execution provenance for LLM agents [15], and separate work argues for preserving negative ML results rather than selective omission [14]. Preregistration, FAIR principles, and nanopublication-style lineage further support reproducibility [5–7]. HydraDG integrates rather than replaces these foundations: probabilistic outputs, deterministic transforms, malformed cells, failures, abstentions, blocked states, and underpowered terminations remain typed evidence with explicit claim-promotion ceilings enforced via FCO/FCG handoff receipts. Companion custody-framework preprints (deferred to camera-ready de-anonymization) define FCO/FCG formally; they establish framework provenance, not the HydraDG empirical results reported here.

3 Framework

3.1 Mechanical Scientific Models

We propose the term *Mechanical Scientific Model* for a computational scientific system whose internal evidence-state transitions are mechanically exposable rather than inferred post hoc from model

84 prose. We do not claim that this term is already established in the literature; it names a proposed
85 model class whose required properties can be checked against an implementation.

86 Let S_t denote the governed evidence-and-claim state at step t . A Mechanical Scientific Model
87 advances by governed transforms

$$S_t \xrightarrow[E_t]{T_t} S_{t+1},$$

88 where T_t is a versioned, auditable transform (deterministic scorer, ingest pass, governed model call,
89 or tool invocation) and E_t is the evidence bound to that transition. Every transition must emit or
90 bind:

- 91 1. source or upstream evidence;
- 92 2. transform, tool, or model identity (with deterministic vs. probabilistic output separation);
- 93 3. derived evidence or parser state;
- 94 4. a bounded claim with an explicit ceiling; and
- 95 5. an artifact or state delta recorded as successor state rather than silent history rewriting.

96 Required properties include: explicit evidence state; governed and versioned state transitions; deter-
97 ministic vs. probabilistic output separation; preservation of positive, null, negative, failed, blocked,
98 timeout, abstention, and contradictory outcomes; explicit claim ceilings; forward and reverse ev-
99 idence traceability; and successor-state recording. Figure 1 situates the Mechanical Scientific
100 Method, FCO/FCG custody, the proposed model class, HydraDG as the evaluated implementation,
101 Vithia as a companion exemplar, and execution components used in this submission.

102 3.2 Custody objects

103 An **FCO** (*Fractal Custody Object*) materializes one bounded transformation: e.g., a model call,
104 scorer pass, or ingest operation. An **FCG** (*Fractal Custody Graph*) append records directed edges
105 between FCOs with hash-linked roots [11]. FCO/FCG provide the custody substrate on which
106 Mechanical Scientific Models record transitions; HydraDG is one implementation that preserves
107 failures, nulls, and malformed outputs in custody (a failure-complete *behavior*, not a redefinition of
108 FCO/FCG). Claim ceilings (exploratory mechanistic falsification, descriptive only, etc.) are declared
109 at preregistration and enforced at promotion time.

110 3.3 Anticube evidence typing

111 Portfolio anticube semantics classify evidence along established vs. exploratory and positive vs.
112 negative axes rather than a single success bit. In this checkout, Antigence-derived anticube state
113 appears in supplement figures F11–F12 and table A7_ANTICUBE_SOT_STATE_LEDGER; temporal
114 trajectory coordinates are sparse and labeled NOT_IN_CHECKOUT where source graphs are absent.
115 These panels document classification mechanics only; they do not upgrade EXP-008/009 claim
116 ceilings.

117 3.4 Experimental freeze

118 Each experiment declares:

- 119 • frozen case manifest and prompt-contract hashes;
- 120 • conditions (e.g., flat prose vs. structured FCG);
- 121 • models with expected runtime digests;
- 122 • primary endpoint and aggregation rule;
- 123 • explicit gates for confirmatory vs. exploratory claims.

124 Execution emits START_RECEIPT, per-cell raw outputs with prompt/response SHA-256, parser state,
125 and terminal receipts. Integrity failure (duplicate keys, missing dependencies, silent model substi-
126 tution) blocks promotion; scientific failure (timeout, abstention, malformed JSON) is retained as
127 terminal evidence.

F1: Proposed hierarchy (conceptual; not treatment-effect evidence)

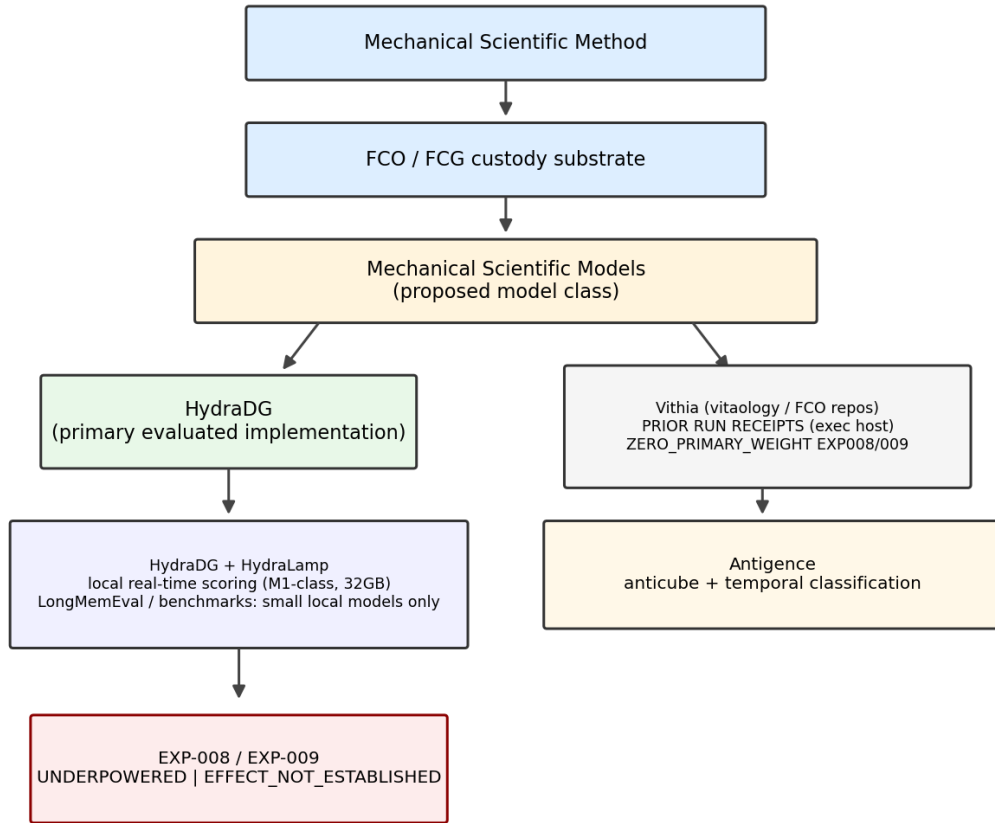


Figure 1: Proposed intellectual hierarchy (conceptual; not empirical effect evidence). HydraDG is the primary experimentally evaluated implementation; Vithia maps to companion repository vitaology (runs retained on the scientific execution host, NOT_IN_CHECKOUT here) and fractal-custody-objects; Antigence anticube panels are supplement-only; EXP-008/009 remain *underpowered*.

128 3.5 Governed local execution

129 Where preregistered, local model calls route through a governed bridge that records selection age,
 130 swap posture, and resolved digest. Direct runtime invocation and governed invocation are separate
 131 evidence classes; one is not silently labeled as the other. EXP-008/009 executed on an Apple Sili-
 132 con Mac Studio class workstation (32 GB RAM) with HydraDG scoring and HydraLamp systems-
 133 validation receipts on the same host class.

134 3.6 Evaluation model policy

135 All preregistered scientific endpoints in this submission use small local open-weights models only
 136 (qwen3:1.7b, qwen2.5-coder:7b; LongMemEval historical lanes per A4 matrix). Frontier cod-
 137 ing agents assisted manuscript preparation only; they did not execute frozen primary scorers or
 138 replace verdict-bound cells.

Table 1: Terminal preregistered studies (HydraDG IC Failure Learning). *Underpowered* denotes insufficient paired evidence for confirmatory promotion, not proof of null effect.

Study	Primary verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

4 Experimental Setup

4.1 Task family

The IC Failure Learning suite uses synthetic agent-failure vignettes with preregistered endpoints such as E06 (prevents exposure of out-of-vault media). Cases are drawn from a frozen CASES.jsonl manifest; each case receives multiple replicates per condition.

4.2 Conditions and models

C0 (FLAT_PROSE): failure-handling guidance as unstructured text.

C1 (STRUCTURED_FCG): the same semantic content composed through a frozen FCG retrieval contract.

EXP-008 and EXP-009 used local models qwen3:1.7b and qwen2.5-coder:7b under identical case and scorer contracts on private M1-class hardware (32 GB RAM). Thinking/reasoning modes were held consistent with preregistration. Primary aggregation: majority of three replicates per case. Historical LongMemEval track-03 retrieval ablations (supplement F4–F5) replay the same local-only policy with qwen2.5-7b and deepseek-r1-14b; they are labeled HISTORICAL_SUPPORTING_ZERO_PRIMARY_WEIGHT and do not promote primary claims here.

4.3 AI and agent methodology disclosure

Frontier coding agents assisted with repository tooling and manuscript preparation; they are not listed as authors. All scientific endpoints come from preregistered local model executions with hashed prompts and responses. Routine editing does not constitute a reported experimental intervention.

4.4 Power and claim gates

E06 endpoints are known limited-power under the frozen case set. Confirmatory ordering claims require preregistered discordant pairs; exploratory secondary statistics are quarantined from primary promotion.

5 Results

Table 1 summarizes terminal verdicts for preregistered studies. Figure 2 decomposes the same receipts into attrition counts and data_quality rates ($n_{\text{raw}} = 300$ per study). Both experiments executed the full raw matrix with complete custody append.

EXP-008. Structured FCG retrieval did not yield promotable confirmatory evidence on E06 under the frozen design ($n = 2$ paired cases per model stratum at aggregation scope). Non-zero malformed and abstention rates are preserved in custody.

EXP-009. Primary verdict remains UNDERPOWERED for confirmatory E06 ordering. An exploratory secondary pattern was recorded as directionally positive at the descriptive layer only; it is **not promoted** to a causal claim.

Stage-2 baseline. Prior IC Failure Learning stage-2 execution (414 scored rows across two models) established FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED, providing historical context for the structured-retrieval falsification lane. LongMemEval track-03 historical lanes (local qwen2.5-7b / deepseek-r1-14b, frozen primary scorer) are retained in the supplement for

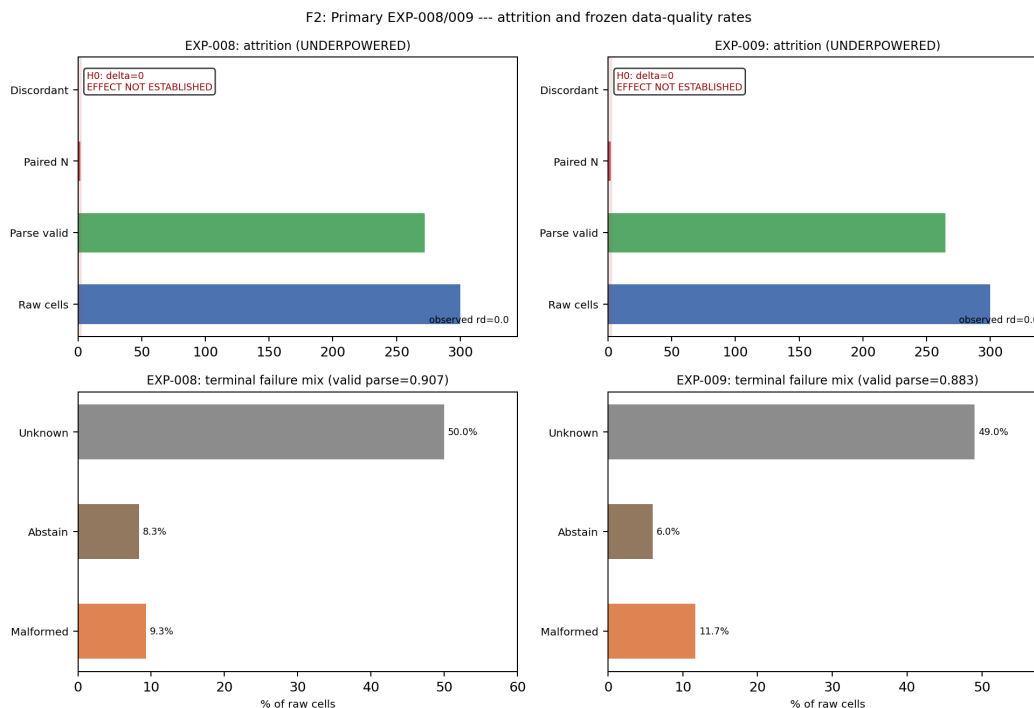


Figure 2: Primary EXP-008/009 outcomes from frozen verdict JSON: top panels show attrition (raw → paired); bottom panels show terminal failure rates (malformed, abstain, unknown). H_0 reference shown; UNDERPOWERED — effect not established (not proof of null).

Table 2: Systems-validation outcomes (custody mechanics only; not treatment-effect evidence).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

177 audit only; they were not executed with frontier models and do not upgrade primary EXP-008/009
 178 claims.

179 5.1 Failure-preserving systems validation

180 The preregistered retrieval experiments above test whether structured context improves a frozen end-
 181 point. A separate systems-validation suite asks a different question: does the custody mechanism
 182 preserve integrity under perturbation, tampering, concurrency, replay, and external execution fail-
 183 ure? Table 2 summarizes source-verified outcomes from a frozen HydraLamp perturbation matrix;
 184 these results evaluate custody mechanics and do **not** constitute evidence that structured FCG context
 185 improves the EXP-008/009 endpoint.

186 The tamper suite used synthetic cases only (not real security incidents). In a live provider repair
 187 ladder, earlier infrastructure gates passed before a structured call hit an external quota limit; the
 188 failure remained a terminal auditable state rather than being erased or substituted. The same cus-
 189 tody primitives have been instantiated in sibling systems for local-model orchestration, determinis-
 190 tic scientific-document atomization, offline anomaly detection (Antigenic anticube lanes), bounded
 191 bioinformatics, substrate generation, and operator-facing scientific workflow monitoring; these im-

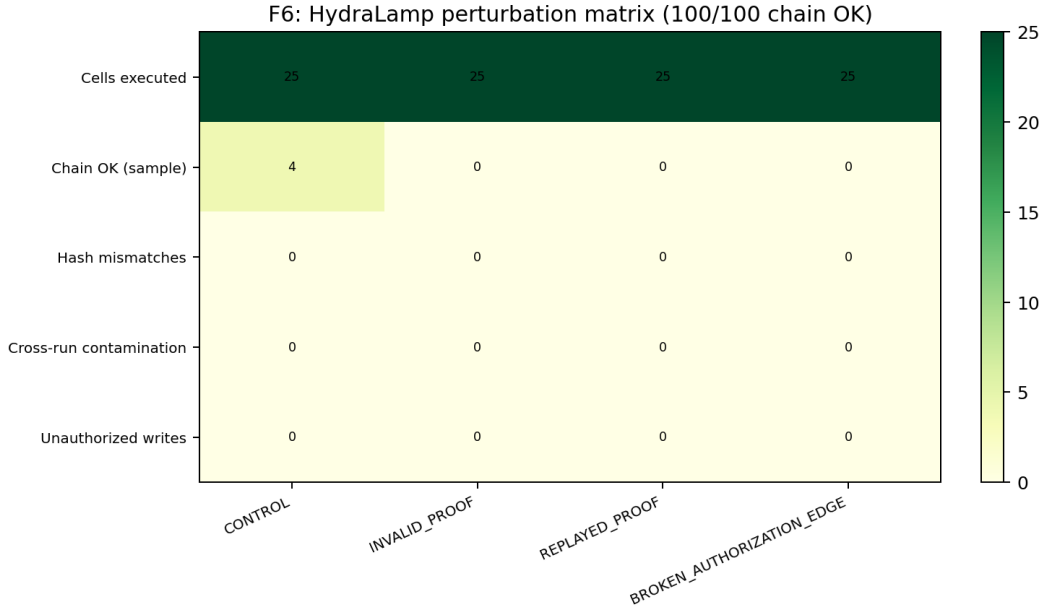


Figure 3: HydraLamp perturbation test matrix (systems robustness only; not treatment-effect evidence).

192 plementations motivate transfer studies but do not provide additional evidence for the treatment
 193 effect reported here.

194 6 Discussion

195 The principal contribution of this submission is not evidence that structured FCG retrieval improves
 196 failure prevention. EXP-008 and EXP-009 remain *underpowered*; the empirical retrieval effect on
 197 the preregistered E06 endpoint is *not established*. The broader methodological proposition is that
 198 scientific computational models can expose their evidence-state transitions mechanically—through
 199 governed receipts, explicit claim ceilings, and preserved terminal states—so that null, failure, and
 200 blocked outcomes remain part of model state rather than disappearing from the public record.

201 HydraDG demonstrates bounded custody and systems-validation behavior under that contract: fail-
 202 ures are first-class, hashes are mandatory, and claim ceilings block over-interpretation. HydraLamp
 203 perturbation, tamper, concurrency, and replay results (Table 2) evaluate custody mechanics only.
 204 Cross-repo census row `vitaology | EXP-001. .014` is admitted as `RELATED_IMPLEMENTATION`
 205 at `ZERO_PRIMARY_WEIGHT`; `vitaology` run receipts are retained on the scientific execution host and
 206 are not vendored into this submission checkout (`NOT_IN_CHECKOUT`). Antigence anticube panels
 207 (F11–F12) type evidence for audit; they are not confirmatory treatment-effect evidence for EXP-
 208 008/009. The EXP-008/009 underpowered terminations bound what cannot yet be claimed while
 209 preserving malformed and timeout cells for audit.

210 6.1 Limitations

211 Limitations include bounded case replication, deliberate local-only small-model lanes in the reported
 212 matrix (no frontier API substitution), and absence of runtime-equivalent successor replication in pri-
 213 mary results. Execution hardware was constrained to M1-class Apple Silicon with 32 GB RAM;
 214 results do not establish performance on datacenter GPU fleets or frontier-model endpoints. A pre-
 215 registered Qwen 3.8 successor lane exists but is non-terminal and omitted from primary results.
 216 Whole-project hierarchical atomization (SeedGraph v1a validation) ran for multiple days but was
 217 interrupted during final Parquet serialization without a `BUILD_RECEIPT`; partial node/edge artifacts
 218 are not readback-safe, so terminal whole-corpus atomization is not established in this submission.

219 Biological, protein-structure, and cellular perturbation applications of the custody framework remain
220 future validation targets, not demonstrated generalizations of the agent-context results here.

221 6.2 Future directions

222 **Agent systems.** Larger successor replications, governed multi-agent handoffs, selective replay, and
223 resource-aware custody gates under preregistered manifests.

224 **Structured retrieval at scale.** Checkpointed ingest and transactionally durable atomization so long
225 hierarchy builds can resume from batch boundaries rather than materializing the entire graph at
226 finalization.

227 **Cross-project federation.** Project-local FCG roots with explicit cross-project references and no
228 automatic evidence admission between repositories.

229 **Cryptographic custody.** Authorized signatures and MMR commitments where policy permits;
230 hashes remain identity commitments unless a verified signing operation occurred.

231 All directions above are *planned* unless independently measured; they do not extend the EXP-
232 008/009 empirical claims.

233 7 Conclusion

234 We proposed the term *Mechanical Scientific Model* for a computational model class whose gov-
235 erned state transitions expose evidence, separate deterministic from probabilistic outputs, preserve
236 failure-complete terminal states, and bind explicit claim ceilings with forward and reverse traceabil-
237 ity. FCO/FCG provide the custody substrate; HydraDG is one implementation examined here under
238 that proposal. Two preregistered HydraDG studies (EXP-008, EXP-009) closed as *underpowered*,
239 demonstrating bounded custody and systems behavior while withholding confirmatory promotion
240 of structured retrieval effects on failure-prevention endpoints. We recommend explicit Mechanical
241 Scientific Method discipline—custody receipts, claim ceilings, and terminal-state preservation—as
242 baseline infrastructure for NewInML-style agent evaluations, without claiming that structured re-
243 trieval efficacy has been established.

244 References

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A Full EXP-008 / EXP-009 statistical audit

Table 3 records terminal verdicts with parse rates reverse-traced to frozen verdict JSON receipts. p -values remain *not informative* under the preregistered underpowered design.

Table 3: Terminal preregistered study audit (appendix).

Study	Verdict	Raw cells	Valid parse rate
EXP-008	UNDERPOWERED	300	0.907
EXP-009	UNDERPOWERED	300	0.883

B Stage-2 M0/M1/M2 baseline

Stage-2 IC Failure Learning execution (414 scored rows across two models) closed as FAILURE_LEARNING_BEHAVIOR_IMPROVEMENT_NOT_ESTABLISHED; descriptive only.

C HydraLamp systems-validation matrix

See Table 4 for custody-mechanics outcomes (not treatment-effect evidence).

Table 4: Systems-validation matrix (appendix).

Validation	Scope	Outcome
Perturbation matrix	100 cells	100/100 chain verification
Synthetic tamper suite	8 modes	8/8 detected
Concurrent execution	10 runs	PASS (10 unique run IDs)
Replay/restart recovery	44 events	PASS
Live provider ladder	R0–R6	Bounded external failure preserved

287 **D Deterministic verification and tooling**

288 Figure 4 and DETERMINISTIC_AUDIT_TOOLING_LEDGER.jsonl document the custody verification
289 pipeline with frozen script SHA-256 and R1/R2/R3 roots.

Source bytes → SHA-256 → Atom → ResultAtom → Claim/Table → FCG delta → exact-SHA verification

Figure 4: FIG-001: governed custody pipeline from source bytes to exact-SHA verification (structured data; self-generated).

290 **E Citation and reference custody**

291 Portfolio recovery bounded 185 occurrences to unique identities; only proposition-backed references
292 are admitted. Citation fabrication literature informs desk-rejection risk [16], [17], [13], [18], [12].

293 **F Requirement, template, and deadline drift**

294 The NewInML workshop submission form requests a CC BY 4.0 article license for this non-archival
295 track; OpenReview platform terms govern platform use separately from article and repository li-
296 censes. Repository code remains Apache-2.0 (separate layer).

297 **G Prior-art comparison matrix**

298 Related-work boundaries include PROV-O, CWLProv, and Workflow Run RO-Crate [16], [17], [13],
299 [18], [12]. Prior-art search responses remain *DISCOVERY_ONLY*; not promoted to verified empiri-
300 cal novelty proof.

301 **H SeedGraph / FCO / FCG custody audit**

302 TOTAL_VERIFIED_INGEST_COMPLETE=NO with verified coverage 31.55% (307/973).

303 **I Planned and nonterminal lanes**

304 GPU SGLang canary, Q38 XENV, and full verified ingest remain nonterminal. All appendix cita-
305 tions resolve through the single main-text bibliography authority.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

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Justification: Experimental Setup and Results sections disclose frozen manifests, conditions, models, scorers, and terminal verdict receipts referenced as internal anonymized artifacts.

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